Circular motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate inward net force to a change in velocity direction.

SUCCESS CRITERIA - I can...

1. Draw tangent velocity and inward acceleration.
2. Use v^2/r or a supplied acceleration.
3. Identify the real interaction providing centripetal force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “centripetal” and “radius” to explain your model or result. Sentence starters are optional.

centripetal: directed toward the center of a circular path

radius: distance from the center to the edge of a circle or sphere

net force: the vector sum of all forces on one object

01 Bring back an earlier idea

Two balls launch horizontally from the same height at 3 and 15 m/s. Ignore air resistance.
Compare fall times and horizontal ranges.

Word help: horizontal = along a level line, usually the x direction

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $a_c = v^2/r$; $F_{net} = mv^2/r$.
c means centripetal: toward the center. This is the net inward force, not an extra force.

Worked reasoning: A 2 kg object moves at 3 m/s in a circle of radius 3 m. Inward acceleration is $v^2/r=3$ m/s²; net force is 6 N inward.

Your turn: Draw tangent velocity and inward net force at two different points.

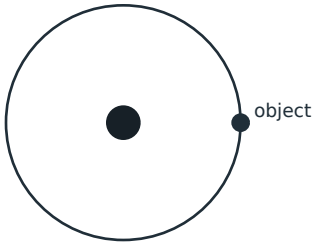
Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

03 Look and explain

Add velocity and acceleration arrows to the marked point on the circle.
Name one interaction that could provide the inward force.

Sentence starter: The ... in my model represents ... . Its label shows ... .



The center is the attracting body; the small dot on the circle is the orbiting object.

Marzano check - circle: ① ② ③

04 Build the solution

A 2 kg object has supplied inward acceleration 3 m/s².
Find inward net force. If acceleration doubles, what happens to force?

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

A string breaks while a ball moves in a circle. Neglect other forces over the next instant.
Sketch its initial path after release.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: 1 2 3

06 Choose a method

A 8 kg cart starts from rest under 80 N net force right for 7 s.
Find acceleration and final velocity.
State one assumption.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

07 Exit check

A car turns at steady speed. Can net force be zero? **Explain** using velocity.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Return to the worked example in problem 2. At the same radius and mass, compare forces for speeds 3 and 6 m/s. Why is the answer not just twice as much?

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3